

Model Explanation

- Purpose: The State Machine Diagram is used to model the lifecycle of the Order Object. It focuses on the various "States" an order transitions through, ensuring the system handles every possible status change correctly without data loss.
- How it Connects:
 - Connection to Activity Diagram: The "States" in this model are the direct results of the "Actions" performed in the System and Driver swimlanes of the Activity Diagram.
 - Connection to Non-Functional Requirements: It ensures Data Consistency, as it provides a single source of truth for the order status across the Customer, Restaurant, and Driver applications.

Order Lifecycle States:

- Pending: The initial state when a user adds items to the cart but hasn't checked out.
- Placed: Transition occurs once the payment is successfully verified.
- Preparing: State enters when the restaurant accepts the order via their dashboard.
- Ready for Pickup: The kitchen marks the food as ready.
- Out for Delivery: Transition occurs when the assigned driver picks up the order.
- Delivered: The final state once the driver confirms the drop-off.
- Cancelled: A terminal state that can be reached if the user cancels (before preparation) or the restaurant rejects the order.